

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} * \begin{bmatrix} w_{11} & w_{12} \\ w_{21} & w_{22} \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix}$$

$$y_{11} = w_{11} \cdot x_{11} + w_{12} \cdot x_{12} + w_{21} \cdot x_{21} + w_{22} \cdot x_{22}$$

$$y_{12} = w_{11} \cdot x_{12} + w_{12} \cdot x_{13} + w_{21} \cdot x_{22} + w_{22} \cdot x_{23}$$

$$y_{21} = w_{11} \cdot x_{21} + w_{12} \cdot x_{22} + w_{21} \cdot x_{31} + w_{22} \cdot x_{32}$$

$$y_{22} = w_{11} \cdot x_{22} + w_{12} \cdot x_{23} + w_{21} \cdot x_{32} + w_{22} \cdot x_{33}$$

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial y_{11}} \cdot \frac{\partial y_{11}}{\partial w_{11}} + \frac{\partial L}{\partial y_{12}} \cdot \frac{\partial y_{12}}{\partial w_{11}} + \frac{\partial L}{\partial y_{21}} \cdot \frac{\partial y_{21}}{\partial w_{11}} + \frac{\partial L}{\partial y_{22}} \cdot \frac{\partial y_{22}}{\partial w_{11}}$$

$\downarrow$ x_{11} $\downarrow$ x_{12} $\downarrow$ x_{21} $\downarrow$ x_{22}

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix}$$

$$\frac{\partial L}{\partial w_{12}} = \frac{\partial L}{\partial y_{11}} \cdot \frac{\partial y_{11}}{\partial w_{12}} + \frac{\partial L}{\partial y_{12}} \cdot \frac{\partial y_{12}}{\partial w_{12}} + \frac{\partial L}{\partial y_{21}} \cdot \frac{\partial y_{21}}{\partial w_{12}} + \frac{\partial L}{\partial y_{22}} \cdot \frac{\partial y_{22}}{\partial w_{12}}$$

$\downarrow$ x_{12} $\downarrow$ x_{13} $\downarrow$ x_{22} $\downarrow$ x_{23}

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix}$$

$$\frac{\partial L}{\partial w_{21}} \rightarrow \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix}$$

$$\frac{\partial L}{\partial w_{22}} \rightarrow \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \frac{\partial L}{\partial w_{11}} & \frac{\partial L}{\partial w_{12}} \\ \frac{\partial L}{\partial w_{21}} & \frac{\partial L}{\partial w_{22}} \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} * \begin{bmatrix} \frac{\partial L}{\partial y_{11}} & \frac{\partial L}{\partial y_{12}} \\ \frac{\partial L}{\partial y_{21}} & \frac{\partial L}{\partial y_{22}} \end{bmatrix}$$

$$y_1 = w_{11} \cdot x_1 + w_{12} \cdot x_2 + w_{21} \cdot x_3 + w_{22} \cdot x_4$$

$$y_{12} = w_{11} \cdot x_2 + w_{12} \cdot x_{13} + w_{21} \cdot x_{22} + w_{22} \cdot x_{23}$$

$$y_{21} = w_{11} \cdot x_{21} + w_{12} \cdot x_{22} + w_{21} \cdot x_{31} + w_{22} \cdot x_{32}$$

$$y_{22} = w_{11} \cdot x_{22} + w_{12} \cdot x_{23} + w_{21} \cdot x_{32} + w_{22} \cdot x_{33}$$

$$\frac{\partial L}{\partial x_1} = \frac{\partial L}{\partial y_{11}} \cdot \frac{\partial y_{11}}{\partial x_1} + \frac{\partial L}{\partial y_{12}} \cdot \frac{\partial y_{12}}{\partial x_1} + \frac{\partial L}{\partial y_{21}} \cdot \frac{\partial y_{21}}{\partial x_1} + \frac{\partial L}{\partial y_{22}} \cdot \frac{\partial y_{22}}{\partial x_1}$$

$$= \begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow \\ w_{11} & 0 & 0 & 0 \end{matrix}$$

$$\frac{\partial L}{\partial x_{12}} = \frac{\partial L}{\partial y_{11}} \cdot \frac{\partial y_{11}}{\partial x_{12}} + \frac{\partial L}{\partial y_{12}} \cdot \frac{\partial y_{12}}{\partial x_{12}} + \frac{\partial L}{\partial y_{21}} \cdot \frac{\partial y_{21}}{\partial x_{12}} + \frac{\partial L}{\partial y_{22}} \cdot \frac{\partial y_{22}}{\partial x_{12}}$$

$$= \begin{matrix} \downarrow & \downarrow & \downarrow & \downarrow \\ w_{12} & w_{11} & 0 & 0 \end{matrix}$$

$$\frac{\partial L}{\partial x_{13}} = \begin{matrix} 0 & w_{12} & 0 & 0 \end{matrix}$$

$$\frac{\partial L}{\partial x_{21}} = \begin{matrix} w_{21} & 0 & w_{11} & 0 \end{matrix}$$

$$\frac{\partial L}{\partial x_{22}} = \begin{matrix} w_{22} & w_{21} & w_{12} & w_{11} \end{matrix}$$

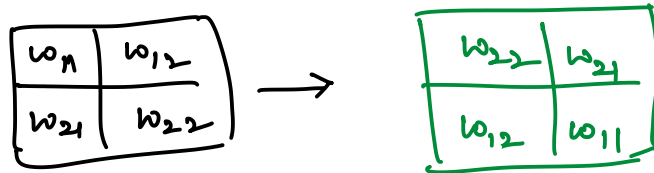
$$\frac{\partial L}{\partial x_{23}} = \begin{matrix} 0 & w_{22} & 0 & w_{12} \end{matrix}$$

$$\frac{\partial L}{\partial x_{31}} = \begin{matrix} 0 & 0 & w_{21} & 0 \end{matrix}$$

$$\frac{\partial L}{\partial x_{32}} = \begin{matrix} 0 & 0 & w_{22} & w_{21} \end{matrix}$$

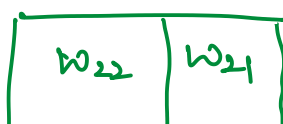
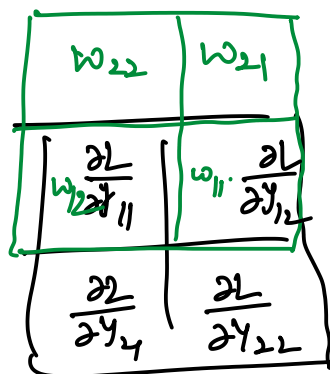
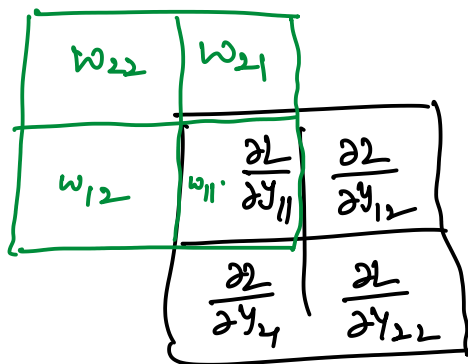
$$\frac{\partial L}{\partial x_{33}} = \begin{matrix} & & & w_{22} \end{matrix}$$

→ Take the 180° rotated version of the filter/weights



equivalent to taking a vertical flip then a horizontal flip

→ Now take a 'full' convolution of the loss gradients w.r.t this flipped filter



$\frac{\partial L}{\partial y_{11}}$	$w_{21} \frac{\partial L}{\partial y_{12}}$	w_{11}
$\frac{\partial L}{\partial y_{21}}$	$\frac{\partial L}{\partial y_{22}}$	

w_{22}	$w_{21} \frac{\partial L}{\partial y_{11}}$	$\frac{\partial L}{\partial y_{12}}$
w_{12}	$w_{11} \frac{\partial L}{\partial y_{21}}$	$\frac{\partial L}{\partial y_{22}}$

one may proceed all the way to the
bottom right-